

Switches  
 ↳ Connect / disconnected

\* type

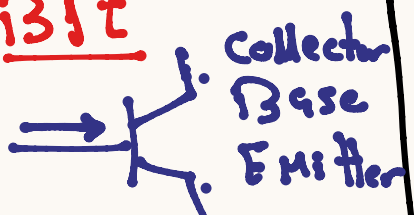
\* Mechanical

\* electrical switch

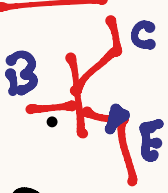
↳ transistor

↳ Relay

\* Bit



NPN



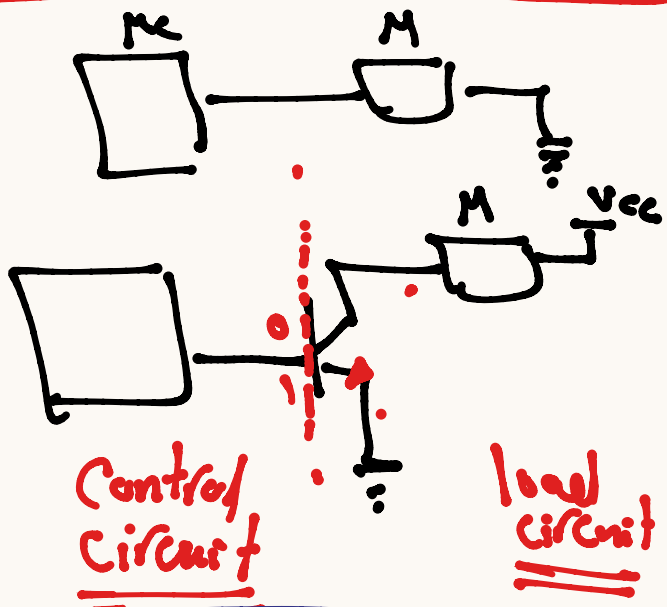
$B=0$   
 $\underline{O.C.}$   
 $B=1$   
 $\underline{S.C.}$

PNP

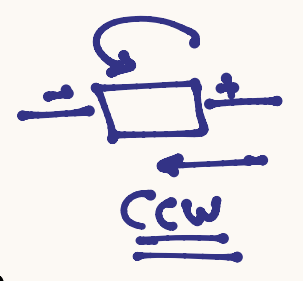
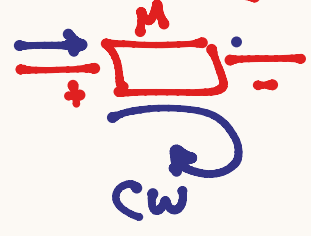


$B=0$   
 $\underline{S.C.}$   
 $B=1$   
 $\underline{O.C.}$

Mosfet



DC Motor



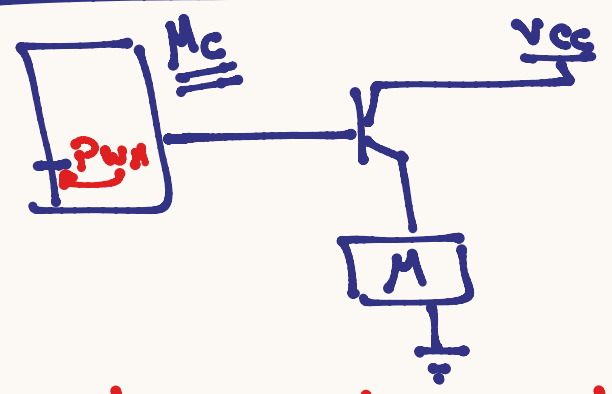
\* → on / off only

\* → on / off / direction  
 off  
 CW  
 CCW

\* → on / off / speed. (PWM)

→ on / off / Direction / speed.

Motor on / off Control



\* Motor Direction Control

↳ need driver

↓  
 Base on H-bridge concept

H-bridge

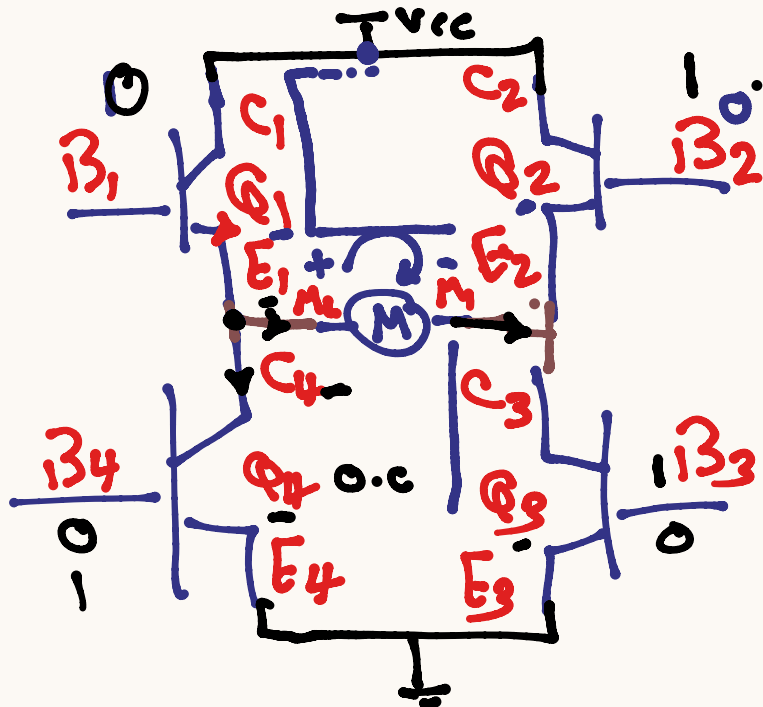
↳ transistor

↳ Relay

4NPN

4PNP

2NPN  
2PNP

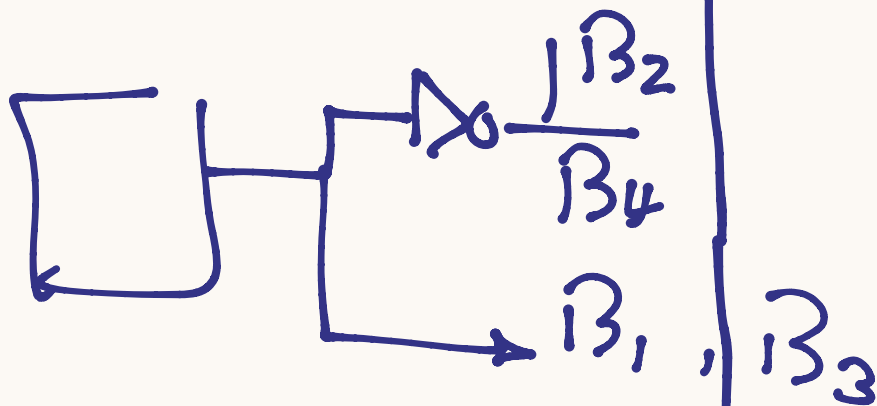


$2NPN \rightarrow B=0 \rightarrow o.c$   
 $B=1 \rightarrow s.c$

| CW        | CEW       | o.p       |
|-----------|-----------|-----------|
| $B_1 = 1$ | $B_1 = 0$ | $B_1 = 0$ |
| $B_2 = 0$ | $B_2 = 1$ | $B_2 = 0$ |
| $B_3 = 1$ | $B_3 = 0$ | $B_3 = 0$ |
| $B_4 = 0$ | $B_4 = 1$ | $B_4 = 0$ |

$B_1 \rightarrow B_3 \rightarrow A_1$

$B_2 \rightarrow B_4 \rightarrow A_2$



## \* Connection

- $C_1, C_2 \rightarrow V_{cc} \rightarrow 5V$
- $E_3, E_4 \rightarrow gnd.$
- $E_1, C_4 \rightarrow M_2$
- $E_2, C_3 \rightarrow M_1$

$B_1 \rightarrow A_1$   
 $B_2 \rightarrow A_2$   
 $B_3 \rightarrow A_3$   
 $B_4 \rightarrow A_4$

## optimized

- $B_1 \rightarrow B_3 \rightarrow A_1$
- $B_2 \rightarrow B_4 \rightarrow A_2$

$1 \ 0 \ 0$   
 $0 \ 1 \ 0$

# \* OP to Coupler

